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(Final Term Past Paper)

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Best of luck!

PAST PAPER

Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

To apply Simpson's 1/3 rule, the number of intervals in the following must be

beacuse the simpson's 1/3 rule is only applied on the even numbers that is the number of interval must be even number.

Answer (Please select your correct option)

- 6
- 7
- 9
- 11

Question No : 2 of 52 Marks: 1 (Budgeted Time 1 Min)

In integrating $\int_1^2 x^2 dx$, by dividing the interval into eight equal parts, width of the interval should be

h = (b-a)/n

given: b=2, a=1, n=8

Answer (Please select your correct option)

- 0.125
- 0.250
- 0.500
- 0.625

Question No : 3 of 52 Marks: 1 (Budgeted Time 1 Min)

The minimum interval in which the root of the equation $x^2 - 3x + 1 = 0$ lie is

Answer (Please select your correct option)

- (0, 1)
- (1, 2)
- (2, 4)
- (0, 2)

Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

............lies in the category of iterative method.

Answer (Please select your correct option)

- None of the given choices
- Bracketing Method
- Regula Falsi Method
- Muller's Method

Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

For the equation $x^3 + 3x - 1 = 0$, the root of the equation lies in the interval......

Answer (Please select your correct option)

- (0, 1)
- (1, 2)
- (1, 3)
- (1, 2)

Question No : 6 of 52 Marks: 1 (Budgeted Time 1 Min)

The quantity of error which is present in the statement of the problem itself ,before finding its solution is called

Inherent errors

It is that quantity of error which is present in the statement of the problem itself, before finding its solution. It arises due to the simplified assumptions made in the mathematical modeling of a problem. It can also arise when the data is obtained from certain physical measurements of the parameters of the problem.

Answer (Please select your correct option)

- Inherent error
- Local round off error
- Local Truncation error
- Typing error

Question No : 7 of 52 Marks: 1 (Budgeted Time 1 Min)

In Regula Falsi Method two points x_n and x_{n+1} are chosen such that f(x_n) and f(x_{n+1}) have -----------signs

Answer (Please select your correct option)

+ve

-ve

Opposite

lec 4

Same

Question No : 8 of 52 Marks: 1 (Budgeted Time 1 Min)

Regula Falsi Method lies in the category of ----------------

Answer (Please select your correct option)

- Iterative method
- Bracketing method
- Random method
- Graphical method

Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

Secant method converges --------than bisection

Answer (Please select your correct option)

- Faster
- Slower
- Equally
- None of the given choices

Question No : 10 of 52 Marks: 1 (Budgeted Time 1 Min)

Muller's method is a generalization of

Muller's method is a generalization of the secant method, in the sense that it does not require the derivative of the function. It is an iterative method that requires three starting points $(p_0, f(p_0))$, $(p_1, f(p_1))$, and $(p_2, f(p_2))$. A parabola is constructed that passes through the three points; then the quadratic formula is used to find a root of the quadratic for the next approximation.

Answer (Please select your correct option)

- Bisection method
- Iteration method
- Secant method
- Regula Falsi method

Question No : 11 of 52 Marks: 1 (Budgeted Time 1 Min)

Diagonal dominance of a coefficient matrix is strictly checked in

Answer (Please select your correct option)

- Muller's method.
- Bisection method
- Jacobi's method pg 69
- Newton-Raphson method

Question No : 12 of 52 Marks: 1 (Budgeted Time 1 Min)

It can be verified that for matrix $A=\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$

Answer (Please select your correct option)

- $AA^{-1}=I$, $I=$ identity matrix
- $AA^{-1}=D$, $D=$ diagonal matrix
- $AA^{-1}=S$, $S=$ symmetric matrix
- $AA^{-1}=Z$, $Z=$ orthogonal matrix

Question No : 13 of 52 Marks: 1 (Budgeted Time 1 Min)

If $[A]$ is an $n \times n$ real symmetric matrix, its eigen values are real, and there exists an orthogonal matrix $[S]$ such that the diagonal matrix $[D]$ is given by

if $[A]$ is an $n\ x\ n$ real symmetric matrix, its eigen values are real, and there exists an orthogonal matrix $[S]$ such that the diagonal matrix D is
$[S^{-1}][A][S]$

Answer (Please select your correct option)

- $[D]=[S]^{-1}[A][S]$ **pg104**
- $[D]=[S]^{-1}[S][A]$
- $[D]=[S]^{2}[A][S]$
- None of the above

Question No : 14 of 52 Marks: 1 (Budgeted Time 1 Min)

The symbol used for average operator is

Answer (Please select your correct option)

- Δ
- μ pg 265
- ∇
- δ

Question No : 15 of 52

Marks: 1 (Budgeted Time 1 Min)

P in Newton's backward difference formula is defined as

Answer (Please select your correct option)

- $p = (\frac{x - x_0}{h})$
- $p = (\frac{x + x_0}{h})$
- $p = (\frac{x + x_n}{h})$
- $p = (\frac{x - x_n}{h})$

pg 266

Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)

Given the following data

x	1	2	-4
$f(x)$	3	-5	-4

last example of pg 135 of pdf handouts

Which formula is useful in finding the interpolating polynomial?

Answer (Please select your correct option)

- Newton's backward difference formula
- Lagrange's interpolation formula
- None of the given choices
- Newton's forward difference formula

Question No : 17 of 52 | Marks: 1 (Budgeted Time 1 Min)

Newton's divided difference interpolation formula is used when the values of the independent variable are

To use Newton's divided difference interpolation formula, the values of independent variables should not be equally spaced.

Answer (Please select your correct option)

- Equally spaced
- Not equally spaced
- Constant
- None of the above

Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

Given the following data

x	4	5	7	10
$f(x)$	48	100	294	900

uncomplete question

Answer (Please select your correct option)

- 94
- 97
- 194
- 52

Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

Given the following data

x	1	3	7
$f(x)$	2	4	10

uncomplete question

Answer (Please select your correct option)

- Newton's forward difference interpolation formula
- Newton's backward difference interpolation formula
- Lagrange's interpolation formula
- None of the given choices

Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

For the given table of values

x	0.1	0.2	0.3	0.4	0.5	0.6
$f(x)$	0.425	0.475	0.400	0.452	0.525	0.575

using two-point equation the value of $f'(0.2)$ is.........:

no idea because equation is not given to calculate

Answer (Please select your correct option)

- 0.75
- -0.75
- 0.5
- -0.5

Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

If $y(x)$ is approximated by a polynomial $p_n(x)$ of degree n then the error is given by

Answer (Please select your correct option)

- $\varepsilon(x) = y(x) + P_n(x)$
- $\varepsilon(x) = y(x) - P_n(x)$
- $\varepsilon(x) = P_n(x) - y(x)$
- $\varepsilon(x) = y(x) \times P_n(x)$

Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)

Let I denotes the closed interval spanned by $x_0, x_1, x_2, x_3, x_4, x_5, x_6, x_7, \overline{x}$. Then $F(x)$ vanishes ------times in the interval I.

formula is n+2.

Answer (Please select your correct option)

- 6
- 9
- 7
- 8

Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)

If $f(x) = 5x^6 + 6x^5 - 7x^3 - 9x^2 + 4x - 3$, then its------derivative is zero for all x.

Answer (Please select your correct option)

- 4th
- 7th

as per calculated

- 6th
- 5th

Question No : 24 of 52

Marks: 1 (Budgeted Time 1 Min)

From the following table of values:

x	1.00	1.05	1.10	1.15	1.20	1.25	1.30
y	1.0000	1.0247	1.0488	1.0724	1.0954	1.1180	1.1402

Answer (Please select your correct option)

- Forward difference operator
- Backward difference operator
- Central difference operator
- None of the given choices

Question No : 25 of 52 Marks: 1 (Budgeted Time 1 Min)

In Richardson's extrapolation method, the extrapolation process is repeated until accuracy is achieved, this is called extrapolation to the ……………..

Answer (Please select your correct option)

- limit
- function
- arbitrary value of 'h'
- none of given choices

Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

While deriving Simpson's 3/8 rule, we approximate f(x) in the form ....

Answer (Please select your correct option)

- $ax+b$
- ax^2+bx+c
- ax^3+bx^2+cx+d
- $ax^4+bx^3+cx^2+dx+e$

Question No : 27 of 52

Marks: 1 (Budgeted Time 1 Min)

Whe we apply Simpson's 3/8 rule, the number of intervals n must be

To apply Simpson's 3/8 rule, the number of intervals must be multiple of 3.

Answer (Please select your correct option)

- Even
- Odd
- Multiple of 3
- Multiple of 8

Question No : 28 of 52 Marks: 1 (Budgeted Time 1 Min)

To apply Simpson's 1/3 rule, valid number of intervals are.....

because To apply Simpson's 1/3 rule, the number of intervals must be even.

Answer (Please select your correct option)

- 7
- 8 ★
- 5
- 3

Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

In integrating $\int_0^1 e^{2x}dx$ by dividing into eight equal parts, width of the interval should be........

check

Answer (Please select your correct option)

- 0.250
- 0.500
- 0.125
- 0.625

Question No : 30 of 52 Marks: 1 (Budgeted Time 1 Min)

Romberg's integration method is ---------- than Trapezoidal and Simpson's rule.

Romberg's integration method is more accurate to trapezoidal and Simpson's rule

Answer (Please select your correct option)

- none of the given choices
- more accurate
- less accurate
- equally accurate

Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

A fourth order ordinary differential equation can be reduced to a system of four ------- order ordinary differential equations.

bure k sath bura mat bano
, (y)

Answer (Please select your correct option)

- First
- Second
- Third
- Fourth

Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

Given $\frac{dy}{dt} = \frac{y-t}{y+t}$ with the initial condition y=1.02 at t=0.02. Using Euler's method, y at t= 0.04, h=0.02 is

$$y_{m+1} = y_m + hf(t_m, y_m)$$

Answer (Please select your correct option)

- 3.0392
- 2.0392
- 1.0392
- 0.0392

Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

In solving the differential equation

$$y' = x + y \;;\; y(0.1) = 1.1$$

$h = 0.1$, By Euler's method $y(0.2)$ is calculated as

$$y_{m+1} = y_m + hf(t_m, y_m)$$

Answer (Please select your correct option)

- 1.44
- 1.11
- 1.22
- 1.33

Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

To solve the ordinary differential equation

$$3\frac{dy}{dx} + xy^2 = \sin x, y(0) = 5,$$

by Runge-Kutta 2nd order method, you need to rewrite the equation as

no idea

Answer (Please select your correct option)

- $\frac{dy}{dx} = \sin x - xy^2, y(0) = 5$
- $\frac{dy}{dx} = \frac{1}{3}\left(\sin x - xy^2\right), y(0) = 5$
- $\frac{dy}{dx} = \frac{1}{3}\left(-\cos x - \frac{xy^3}{3}\right), y(0) = 5$
- $\frac{dy}{dx} = \frac{1}{3}\sin x, y(0) = 5$

Question No : 35 of 52 Marks: 1 (Budgeted Time 1 Min)

In second order Runge-Kutta method

k_1 is given by

Answer (Please select your correct option)

- $k_1 = hf(x_n, y_n)$
- $k_1 = 2hf(x_n, y_n)$
- $k_1 = 3hf(x_n, y_n)$
- None of the given choices

Question No : 36 of 52

Marks: 1 (Budgeted Time 1 Min)

In fourth order Runge-Kutta method, k_3 is given by

$$k_1 = hf\left(t_n, y_n\right)$$

$$k_2 = hf\left(t_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$$

$$k_3 = hf\left(t_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right)$$

$$k_4 = hf\left(t_n + h, y_n + k_3\right)$$

Answer (Please select your correct option)

- $k_3 = hf(x_n + \frac{h}{3}, y_n + \frac{k_2}{3})$
- $k_3 = hf(x_n - \frac{h}{2}, y_n - \frac{k_2}{2})$
- $k_3 = hf(x_n - \frac{h}{3}, y_n - \frac{k_2}{3})$
- $k_3 = hf(x_n + \frac{h}{2}, y_n + \frac{k_2}{2})$

Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

In fourth order Runge-Kutta method, k_4 is given by

$$k_1 = hf(t_n, y_n)$$

$$k_2 = hf\left(t_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$$

$$k_3 = hf\left(t_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right)$$

$$k_4 = hf(t_n + h, y_n + k_3)$$

Answer (Please select your correct option)

- $k_4 = hf(x_n + 2h, y_n + 2k_3)$
- $k_4 = hf(x_n - h, y_n - k_3)$
- $k_4 = hf(x_n + h, y_n + k_3)$
- None of the given choices

Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

The truncation error in Adam's predictor formula is

$$y_{n+1} = y_n + \frac{h}{24}(55f_n - 59f_{n-1} + 37f_{n-2} - 9f_{n-3}) + \frac{251}{720}h\nabla^4 f_n$$

Alternatively, it can be written as

$$y_{n+1} = y_n + \frac{h}{24}[55y'_n - 59y'_{n-1} + 37y'_{n-2} - 9y'_{n-3}] + \frac{251}{720}h\nabla^4 y'$$

This is known as Adam's predictor formula.

The truncation error is $(251/720)h\nabla^4 y'_n$.

Answer (Please select your correct option)

- $\frac{251}{720}h\nabla^4 y_n{}'$
- $\frac{251}{720}h\nabla^4 y_{n+1}{}'$
- $\frac{19}{720}h\nabla^4 y_{n+1}{}'$
- $\frac{19}{720}h\nabla^4 y_n{}'$

Question No : 39 of 52 Marks: 1 (Budgeted Time 1 Min)

In solving the differential equation

$$y' = x^2 + 2xy \;;\; y(0) = 1$$

$h = 0.1$, By Euler's method $y(0.1)$ is calculated as

coca cola tunnnnn

Answer (Please select your correct option)

1

2

3

4

Question No : 40 of 52

Marks: 1 (Budgeted Time 1 Min)

In solving the differential equation

$$y' = x^2 + 2y \;;\; y(1) = 3$$

$h = 1$, By Euler's method $y(2)$ is calculated as

O kudi menu kehndi. Mainu jooti le de sohneya. Main keya naah goriye :P dakhe bae ,,,enjoy b huna chaiye sath sath

Answer (Please select your correct option)

- 4
- 6
- 8
- 10

Question No : 41 of 52 Marks: 2 (Budgeted Time 4 Min)

State the sufficient condition of convergence of the iterative solution to the exact solution

Answer

A sufficient condition for convergence of the iterative solution to the exact solution is

$|a_{ii}| > \sum_{\substack{j=1 \\ j\neq 1}}^{n} |a_{ij}|, \qquad i = 1, 2, \ldots, n$ When this condition (diagonal dominance) is true, Jacobi's method converges

A sufficient condition for convergence of iterative solution to exact solution is |a1| => (greater or equal) |b1|+|c1| |b2| => (greater or equal) |a2|+|c2| |c3| => (greater or equal) |a3|+|b3| For the system a1x+b1y+c1z=d1, a2x+b2y+c2z=d2, a3x+b3y+c3z=d3 Similarly for the system with more variables we can also construct the same condition

Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

Evaluate the integral

$$\int_3^5 (\log x + 1)dx$$

Using Trapezoidal rule

Take h=1

Answer (Please click here to Add Answer)

X	3	4	5
Y= logx+1	0.6020	0.6989	0.7781

Using trapezoidal rule, taking h=1

$$I = \frac{h}{2}(y_0 + y_3 + 2y_1)$$

$$I = \frac{1}{2}(0.6020 + 0.7781 + 2(0.6989))$$

$$I = \frac{1}{2}(1.3801 + 1.3978)$$

$$I = \frac{2.7779}{2} = 1.38895$$

Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

Write a formula for finding the value of k_1 in Fourth-order R-K method.

Answer (Please click here to Add Answer)

Normal Arial 12 100%

$$k_1 = hf(t_n, y_n)$$

$$k_2 = hf\left(t_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$$

$$k_3 = hf\left(t_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right)$$

$$k_4 = hf(t_n + h, y_n + k_3)$$

Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

Write Adam-Moultan's Predictor formula for finding the solution of a differential equation.

Answer (Please click here to Add Answer)

Normal Arial 12 100%

$$y_{n+1} = y_n + \frac{h}{24}\left[55y'_n - 59y'_{n-1} + 37y'_{n-2} - 9y'_{n-3}\right] + \frac{251}{720}h\nabla^4 y'$$

This is known as Adam's predictor formula.

Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

Obtain numerically the solution of

$$y' = x^2 + 2x + y^2, \; y(0) = 1$$

Using Euler's method to find y at $x = 1$, h=1

Answer (Please click here to Add Answer)

use this formula to calculate:

$$y_{m+1} = y_m + hf(t_m, y_m)$$

Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

If $f(2) = -2.6146$ and $f(3) = 4.7610$, then find the first approximation using the Regula-Falsi method

Answer (Please click here to Add Answer) clear image

Normal Arial 12 100%

$$x_3 = x_2 - \frac{x_2 - x_1}{f(x_2) - f(x_1)} f(x_2)$$

$$x_3 = 3 - \frac{3-2}{4.7610-(-2.6146)}(4.7610)$$

$$x_3 = 3 - \frac{4.7610}{7.3756}$$

$$x_3 = 3 - \frac{4.7610}{7.3756} = 3 - 0.6455 = 2.3545$$

$$x_3 = x_2 - \frac{x_2 - x_1}{f(x_2) - f(x_1)} f(x_2)$$

$$x_3 = 3 - \frac{3-2}{4.7610-(-2.6146)}(4.7610)$$

$$x_3 = 3 - \frac{4.7610}{7.3756}$$

$$x_3 = 3 - \frac{4.7610}{7.3756} = 3 - 0.6455 = 2.3545$$

Question No : 47 of 52 Marks: 3 (Budgeted Time 6 Min)

If, in solving a given differential equation, $y_0 = 1, y_1' = 1.1, y_2' = 1.2, y_3' = 1.3, h = 1$

Then find y_4 by Milne's Predictor formula.

Answer (Please click here to Add Answer)

Normal Arial 12 100%

calculate by using this formula

$$y_4 = y_0 + \frac{4h}{3}(2y_1' - y_2' + 2y_3')$$

$$P: y_{n+1} = y_{n-3} + \frac{4h}{3}(2y'_{n-2} - y'_{n-1} + 2y'_n)$$

$$P: y_4 = y_0 + \frac{4h}{3}(2y'_1 - y'_2 + 2y'_3)$$

$$P: y_4 = 1 + \frac{4(1)}{3}(2(1.1) - 1.2 + 2(1.3))$$

$$P: y_4 = 1 + \frac{4}{3}(2.2 - 1.2 + 2.6)$$

$$P: y_4 = 1 + \frac{4}{3}(3.6)$$

$$P: y_4 = 1 + \frac{14.4}{3}$$

$$P: y_4 = \frac{3 + 14.4}{3} = \frac{17.4}{3} = 5.8$$

Question No : 48 of 52 Marks: 3 (Budgeted Time 6 Min)

Find $F(h)$, using Richardson's extrapolation limit, while finding $y'(0.01)$ to the function $y = 1/x$ with h =0.005.

Answer (Please click here to Add Answe

$$F(h)=\frac{y(x+h)-y(x-h)}{2h}$$

$$F(h)=\frac{\frac{1}{0.01+0.005}-\frac{1}{0.01-0.005}}{2(0.005)}$$

$$F(h)=\frac{\frac{1}{0.015}-\frac{1}{0.005}}{0.01}$$

$$F(h)=\frac{66.66-200}{0.01}$$

$$F(h)=\frac{-133.34}{0.01}=-13334$$

Question No : 49 of 52 Marks: 5 (Budgeted Time 10 Min)

Evaluate the integral

$$\int_0^3 (x^2 + x)dx$$

Using Simpson's 3/8 rule

Answer (Please click here to A

X	0	1	2	3
$Y=(x^2+x)$	0	2	6	12

$$\int_0^3 (x^2 + x)dx = \frac{3}{8}h\left[y_0 + y_3 + 3(y_1 + y_2)\right]$$

$$\int_0^3 (x^2 + x)dx = \frac{3(1)}{8}\left[0 + 12 + 3(2 + 6)\right]$$

$$\int_0^3 (x^2 + x)dx = \frac{3}{8}[12 + 24] = \frac{3}{8}(36) = \frac{108}{8} = 13.5$$

Question No : 50 of 52 Marks: 5 (Budgeted Time 10 Min)

Construct a backward difference table from the following values of x and y.

x	-1	0	1	2	3
y=f(x)	10	2	10	62	80

Answer (Please click here to Add Answer)

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x	Y=f(x)	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
-1	10				
0	2	-8			
1	10	8	16		
2	62	52	44	28	
3	80	18	-34	-78	-106

Question No : 51 of 52 Marks: 5 (Budgeted Time 10 Min)

From the following table of values, construct forward difference table.

x	1.00	1.05	1.10	1.15	1.20
y	1.0000	1.0247	1.0488	1.0724	1.0954

Answer (Please click here to Add Answer)

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x	Y=f(x)	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
1.00	1.000				
1.05	1.0247	0.0247			
1.10	1.0488	0.0241	0.0006		
1.15	1.0724	0.0236	-0.0005	-0.0011	
1.20	1.0954	0.023	-0.0006	-0.0001	0.0000

 Marks: 5 (Budgeted Time 10 Min)

Use Runge-Kutta Method of order four to find the values of k_1, k_2, k_3 *and* k_4 for the initial value problem

$$y' = \frac{1}{2}(2x^3 + y), y(1) = 2 \text{ taking } h = 0.1$$

Answer (Please click here to Add Answer)

Normal Arial 12

$$k_1 = hf(t_n, y_n)$$

$$k_2 = hf\left(t_n + \frac{h}{2}, y_n + \frac{k_1}{2}\right)$$

$$k_3 = hf\left(t_n + \frac{h}{2}, y_n + \frac{k_2}{2}\right)$$

$$k_4 = hf(t_n + h, y_n + k_3)$$

$$here\ f(x,y) = \frac{1}{2}(2x^3 + y), \quad h = 0.1, \quad x_1 = 1, \quad y_1 = 2$$

$$K_1 = hf(x_1, y_1) = 0.1\left(\frac{1}{2}\left(2(1^3 + 2)\right)\right)$$

$$K_1 = hf(x_1, y_1) = 0.1\left(\frac{6}{2}\right) = \frac{0.6}{2} = 0.3$$

$$K_2 = hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_1}{2}\right) = 0.1\left(1 + \frac{0.1}{2}, 2 + \frac{0.3}{2}\right)$$

$$K_2 = hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_1}{2}\right) = 0.1(1.05, 2 + 0.15)$$

$$K_2 = hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_1}{2}\right) = 0.1(1.05, 2.15) = 0.1(1.05 + 2.15) = 0.32$$

$$K_3 = hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_2}{2}\right) = 0.1\left(1 + \frac{0.1}{2}, 2 + \frac{0.32}{2}\right)$$

$$K_3 = hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_2}{2}\right) = 0.1(1.05, 2 + 0.16)$$

$$K_3 = hf\left(x_1 + \frac{h}{2}, y_1 + \frac{k_2}{2}\right) = 0.1(1.05, 2.16) = 0.1(1.05 + 2.16) = 0.321$$

$$K_4 = hf(x_1 + h, y_1 + k_3) = 0.1(1 + 0.1, 2 + 0.321)$$

$$K_4 = hf(x_1 + h, y_1 + k_3) = 0.1(1.1, 2.321) = 0.1(0.1 + 2.21575) = 0.3421$$

Now,

$$y_2 = y_1 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$y_2 = 2 + \frac{1}{6}(0.3 + 2(0.32) + 2(0.321) + 0.3421)$$

$$y_2 = 2 + \frac{1}{6}(0.3 + 0.64 + 0.642 + 0.3421) = 2 + \frac{1.9241}{6} = 2 + 0.320683 = 2.320683$$